



CL1002 Programming Fundamentals	Lab 04 Tasks
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NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

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LAB EXERCISES

Task 1 – Hospital Emergency Card Generator

A hospital has introduced an emergency card system to classify patients based on their age and medical condition. When a patient arrives, the system records their name, age, and condition. If the patient is under 12 years old or above 65 years old, they are automatically given a High Priority card. If the condition is marked as critical, the patient is also assigned a High Priority card regardless of their age. For patients whose conditions are moderate, the system assigns a Medium Priority card, while those with mild conditions are assigned a Low Priority card. The program should generate and display the complete emergency card, including the patient's details and their assigned priority level.

Task 2 – Student Course Registration Validator

At a university, students are allowed to register for courses each semester under specific rules. The maximum number of courses a student can take is five. Some courses require prerequisites, meaning the student must have passed certain subjects before registering. Additionally, two courses that have overlapping time slots cannot be selected together. The system should allow a student to enter their desired courses and then validate the registration. If the registration is invalid, the system must display the reasons, such as exceeding the course limit, missing prerequisites, or time clashes.

Task 3 – Smart Energy Consumption Monitor

A smart home system is being developed to track the daily energy usage of household appliances. Each device, such as a fan, air conditioner, or refrigerator, reports its daily consumption in kilowatt-hours (kWh). The system needs to check whether any device crosses the safe threshold of 10 kWh per day. Devices that consume more than this limit should be flagged as high consumption devices. At the end of the day, the program should generate a summary report that displays the energy consumption of each device and clearly highlights any devices that exceeded the threshold.

Task 4 – Employee Shift Scheduling System

A software company is creating a scheduling system to manage employee working hours. Each employee can work a maximum of 40 hours per week, and no single shift can exceed 8 hours. Shifts can only be assigned if the employee is marked as available on that day. The scheduling system should take employee availability and requested shifts as input, then analyse whether the schedule is valid. If an employee is scheduled for too many hours, assigned a shift longer than 8 hours, or given a shift when unavailable, the system should highlight the scheduling conflicts.

Task 5 – Airline Boarding System

An airline has introduced a boarding system that assigns passengers to groups based on their ticket details. Business class passengers and those traveling with children under the age of five are placed in Group A. Economy class passengers sitting in the front rows, specifically rows 1 to 10, are placed in Group B. Those seated in the middle rows, rows 11 to 20, are placed in Group C. Economy passengers seated in rows 21 and above are placed in Group D. The program should take passenger information such as ticket class, seat row, and whether they are traveling with young children, and then assign them to the appropriate boarding group.

Task 6 – University Exam Seating Planner

A university exam hall must be arranged to minimize the chances of cheating. The rule is that no two students from the same course should be seated next to each other. The hall should be filled row by row in a systematic manner, ensuring that students from different courses are placed alternately. If the number of students from a particular course is too high and separation is not possible, the system should issue a warning that the seating plan cannot fully satisfy the rule. The program should take the list of students with their course names and generate a seating arrangement plan according to these conditions.

Task 7 – Banking Transaction Validator

A banking system needs to process customer transactions securely. Each customer account has an available balance, and withdrawals cannot exceed this balance. Additionally, the bank enforces a daily withdrawal limit of Rs. 50,000. If a customer tries to withdraw more than this amount, the transaction must be declined. If a customer attempts more than three invalid transactions in a single day, the system must automatically block their account for security reasons. The program should simulate multiple transactions on a customer account and validate each transaction according to these rules.

Task 8 – Data Backup Scheduler

An IT company runs automated backups to protect its data. The rule is that a full backup must be taken once every seven days, while incremental backups are taken every day to save storage space. If the system detects corruption in any file, it must immediately trigger a full backup regardless of the weekly schedule. The program should simulate a seven-day backup cycle, allowing the system to record daily backups while ensuring that the rules are followed. It should also handle corruption scenarios and switch to a full backup when required.

Task 9 – Online Examination Proctor

An online examination platform needs to ensure that students remain on the test page throughout the exam. The rules are strict: if the student switches their browser tab once, they receive a warning. A second tab switch results in another warning, and on the third switch, the exam is terminated immediately. If the student does not switch tabs at all, the exam is completed successfully. The program should simulate this monitoring process by analyzing a

sequence of student actions and then output whether the student completed the exam, was warned, or had their exam terminated.

Task 10 – Smart Traffic Signal with Emergency Override

A smart city project is implementing an intelligent traffic signal system that adjusts the green light duration based on the number of vehicles detected at an intersection. If fewer than ten vehicles are waiting, the green light remains on for 15 seconds. If the number of vehicles is between ten and thirty, the green light is extended to 30 seconds. For more than thirty vehicles, the green light lasts 60 seconds. However, if an emergency vehicle such as an ambulance or fire truck is detected, the system must immediately override the normal rules and switch the signal to green. The program should simulate this decision-making process and display the green light duration based on the input conditions.